Lab 1 – Product Description

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1 Introduction

A large disparity exists in this country between the amount of pets currently housed in the United States and those that can afford to receive proper veterinary care. As of March of this year, approximately 94 million homes reported having at least one pet within the household. Out of those 94 million households with one or more pets, only about 5.7 million of those pets are properly insured (Facts + Statistics). Assuming all those households only had one pet, that is still only 6.06% of households in the United States with pets that are properly insured. This is indicative of a greater trend of the poor state of the pet health industry within the US. Not only is it hard for pets to receive proper veterinary care due to expenses and the trend of pets being underinsured, the industry is also hard on the veterinarians working with pets. In fact, the suicide rates for veterinarians are much higher on average than the general population at a 1.6 times increase from average seen in male vets and a 2.4 times increase in likelihood seen amongst female veterinarians (Nett, 2019). This can largely be attributed to compassion fatigue, a condition where the prolonged exposure to suffering and trauma can negatively impact a person's mental state. When someone is tasked to deal with animals which are widely considered cute and hard to see in pain for most people suffering from the worst health conditions and being tasked to help them ease their suffering day-in and day-out, it can plainly be seen how compassion fatigue may develop. Not only is there a lack of proper treatment for pets due to expenses in the veterinary care industry, but there is a lack of proper resources and relief for the toll of the stress of treating pets that affects the veterinarians themselves.

The main goal of this product is to take the reins of these struggles and become a leading source of information and help to both pet owners and veterinarians alike, so only appropriately

it has been titled “Take The Lead.” This product will work in conjunction with pet hospitals, ERs, veterinary offices, and veterinarians to deliver a service that not only provides valuable information and resources to pet parents and vets, but also directly helps fund procedures for pets that need them. To achieve this, Take The Lead will host 30-day donation campaigns to raise money for necessary procedures needed by the pets of its users. These campaigns will be initiated by either guaranteeing a delay of payment with the hospital themselves or involve Take The Lead taking a hit on the upfront cost while waiting for the campaign to finish if the care needed is critical and time-sensitive. Campaigns will be created by users who will share their story, share the necessary information and cost of the care needed (confirmed by Take The Lead staff), and will be shared on the Take The Lead application with a shareable link for the user to spread their campaign elsewhere. At the end of the campaign, the revenue accrued will go directly toward the bill by either a direct payment to the hospital or the funds being repaid to Take The Lead. If the campaign generates excess revenue, that money will be given directly to the user and becomes their liability due to the legal risks of monetary collection that Take The Lead prefers to avoid. The 30-day campaigns will not be the only option a user has for assistance as Take The lead will provide additional information for the user on steps to take before starting a campaign, especially if funding is required immediately. Take The lead also provides a centralized hub for information on veterinary care for both pet parents and veterinarians alike. This will include external links to information on pet care, resources on determining if pet insurance is needed and how to get it, proper questions to ask while trying to receive veterinary care, and information on why veterinary care is necessary and why it is as expensive as it is. In total, Take The Lead aims to solve fundamental issues with the veterinary in lack of easy access to proper information and means to pay for unexpected medical expenses.

2 Product Description

Take The Lead is a website/ application which provides a means for users to discover information about veterinary care and provides resources for covering expenses of pet care. To achieve this, Take The Lead allows users to host 30-day fundraising campaigns on the website where they will share their story, set the goal amount needed for the care of their pet, and provide ways to share the campaign with people on the Take The Lead website as well as a shareable link to the campaign to be used anywhere on the broader internet. On the end of Take The Lead, creating these campaigns will begin a 48 hour process of confirming the bill amount with the listed pet hospital, determining if the bill will need to be paid upfront before the campaign or if the payment can be held off, and finalizing the campaign to be shared to the wider Take The Lead community. These campaigns will be paid through individual donations from Take The Lead users. Revenue generated from these campaigns will go to different places depending on circumstances: for instance, Take The Lead has paid a down payment to the hospital, funds will be paid back to Take The Lead. Instead, the campaign revenue can be recursively paid to the hospital, held off to the end of the campaign, or if there is excess revenue generated be paid out directly to the user who started the campaign. If the campaign fails to meet the minimum amount, payment will need to be made out to the pet hospital or Take The Lead if a down payment was put out from the user. Take The Lead also provides many informational resources on pet care, insurance, expenses, and stresses of the industry for pet parents and veterinarians alike.

2.1 Key Product Features and Capabilities

Take The Lead provides many resources both monetarily and informatively for pet parents and veterinary professionals; This allows users to create 30-day fundraising campaigns to help with veterinary care expenses. Take The Lead can put out down payments for immediately necessary veterinary care while the campaign runs. Accrued revenue from these campaigns will be paid back to the hospital or Take The Lead. Excess revenue will be paid back to the user who started the campaign. Campaigns that fail to meet minimum revenue will become the responsibility of the user to pay back the hospital or Take The Lead.

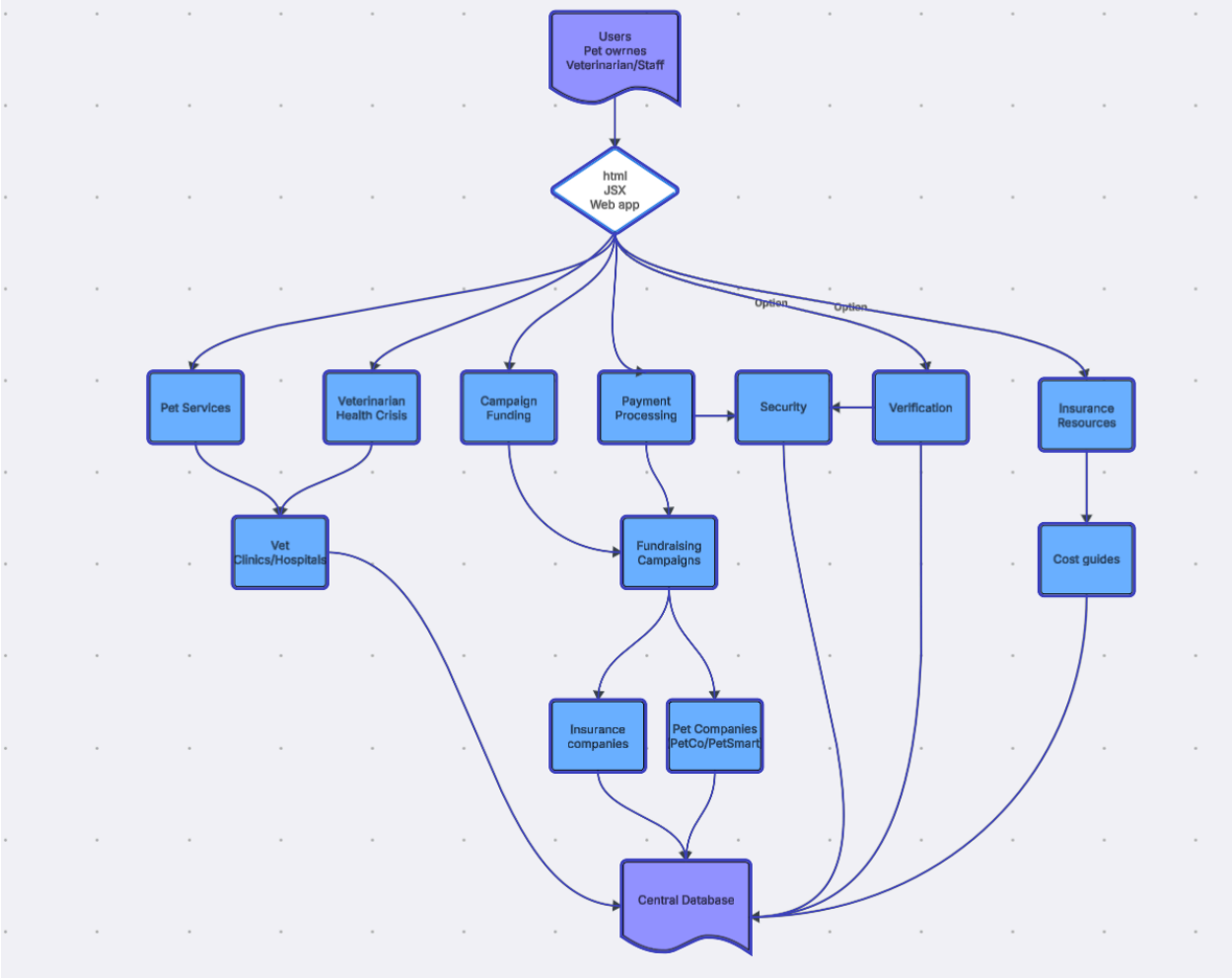
Take the Lead provides resources for pet owners and veterinarians on insurance, care, and help within the industry. Pet parents can research if insurance is necessary for them and have resources to acquire insurance within the Take The Lead website. Pet parents can find information on why veterinary care is important, expensive, and what to ask their veterinarians in relation to their pets. Veterinarians can find resources for help, especially with the high suicide rates of veterinarians access to proper mental health resources is a necessity.

2.2 Major Components (Hardware/Software)

Take The Lead can be run by a user on any modern device that can display webpages with access to the internet (e.g. a personal computer or smartphone). On the back end, Take The Lead will have a central database on a server that hosts all information and sources needed to be displayed on the webpage and record keeping. Software components required include HTML, JSX files, CSS files, and JavaScript.

Figure 1

Major Functional Component Design for Take The Lead



The hardware in the database (shown in Figure 1) will host the necessary information stored in secure files with the webpages for Take The Lead displayed with HTML components and hosted on a server to be shown to users on their devices.

3 Identification of Case Study

This project is designed for two groups: Pet owners and veterinarians. For pet owners, this project is being developed as a way to alleviate the stress of pet injury, illness, and general medical expenses that come with keeping pets. The aim is to help pet owners by providing information and financial aid to contribute to their pet(s)'s health; this is a known group that would benefit from this information as is shown by similar yet not as focused products such as Treasured Memories, The Pet Fund, etc. For veterinarians, it acts as a source of information and a means of alleviating the stresses of their profession; this is a prevalent issue given the heightened rate of suicide amongst veterinarians as outlined by the CDC.

A small group that can test the prototype is a selected few pet owners that ran, or running, or were planning to run similar donation campaigns on other websites such as those listed above. This would provide a sample group that aligns with the target user base while providing feedback on how this product compares to other similar products and in what ways it improves or needs to be improved upon compared to competitors. In the future, the hope is that many underinsured pet owners use this app and give their pets a chance that would have been wasted otherwise.

4 Glossary

Campaign: A short-window online fundraising effort with proceeds contributed toward the necessary care of an animal; contains a story, donation goals, relevant information, and is publicized through the Take The Lead website.

CDC: Short-hand for Center of Disease Control, a national public health and safety agency within the United States.

Member: A user with a registered account of the web application “Take the Lead.”

Preventative Care: Animal care that is to prevent illness. Examples of this are vaccines, check-ups, x-rays of older animals’ joints, etc. This is also termed wellness care.

Story: The 30 day campaign that a Member will run, interchangeable with “campaign.”

TTL: Short-hand for the company Take The Lead

5 References

American Veterinary Medical Association. (n.d.). Home. https://www.avma.org/

Frankie's Friends. (2026). Welcome. https://www.frankiesfriends.org

Insurance Information Institute. (2026). Facts + statistics: Pet ownership and insurance. https://www.iii.org/fact-statistic/facts-statistics-pet-ownership-and-insurance

Nett, R. J., Witte, T. K., Holzbauer, S. M., Elchos, B. L., Campagnolo, E. R., Musgrave, K. J., Carter, K. K., Kurkjian, K. M., Vanicek, C. F., O'Leary, D. R., Pride, K. R., & Funk, R. H. (2019, September 4). Suicide risk for veterinarians and veterinary technicians. NIOSH Science Bulletin, Centers for Disease Control and Prevention. https://www.cdc.gov/niosh/bulletin/2019/suicide-veterinarians.html

Not One More Vet. (2025). Veterinary profession mental wellness. https://www.nomv.org/

The Pet Fund. (2026). Home. https://www.thepetfund.com

Treasured Memories Community Funding. (2019). Treasured memories community funding. https://www.tmcfunding.com